



Memory & Cognition

Lecture - 11



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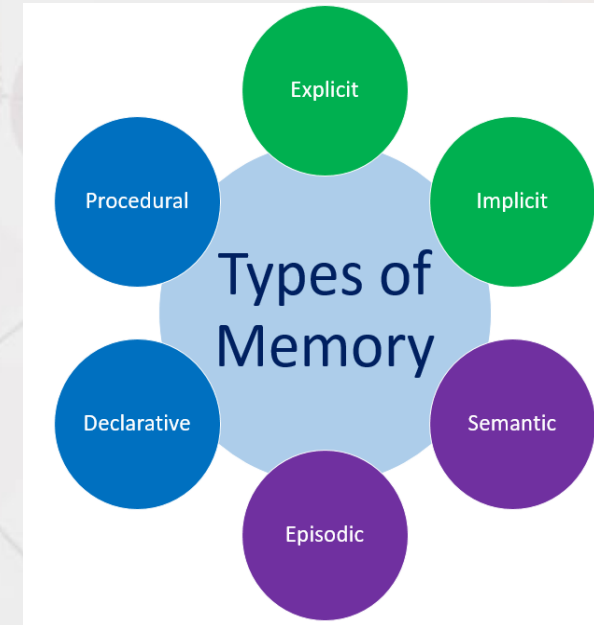
- **What Is Memory?**
- Memory refers to the psychological processes of acquiring, storing, retaining, and later retrieving information.
- Human memory involves the ability to both preserve and recover information.
- Researchers have long believed that memories form due to changes in brain neurons (nerve cells).
- Changes in the connections between nerve cells (synapses) are associated with the learning and retention of new information. Strengthening these connections helps commit information to memory.
- There are three major processes involved in memory: encoding, storage, and retrieval.



Memory Defined

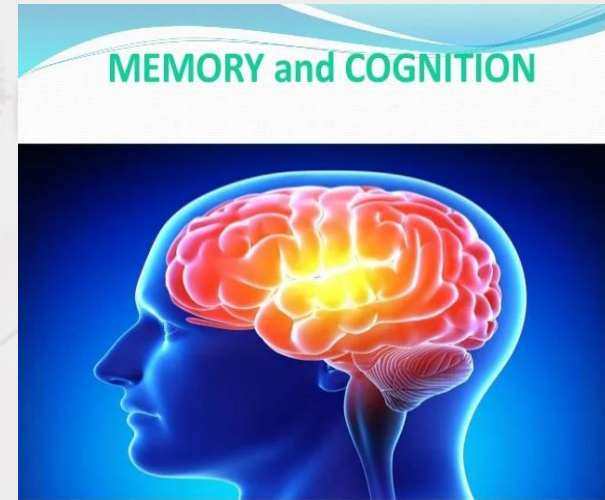


- In order to create a new memory, information must be changed into a usable form, which occurs through a process known as encoding.
- Once the information has been successfully encoded, it must be stored in memory for later use.
- **Our cognitive system for storing and retrieving information**
 - **Encoding or registration**: receiving, processing and combining of received information
 - **Storage**: creation of a permanent record of the encoded information in short term or long term memory
 - **Retrieval, recall or recollection**: calling back the stored information in response to some cue for use in a process or activity





- ✓ **Reviewing** and **rehearsing information** improves the ability to remember it.
- ✓ **Practice strengthens the connections** between the synapses that store that memory.
- ✓ Much of our stored memory lies outside of our awareness most of the time, except when we actually need to use it.
- ✓ The **memory retrieval process** allows us to **bring stored memories** into conscious **awareness**.
- ✓ To use the **information** that has been **encoded** into memory, it first has **to be retrieved**.
- ✓ There are many **factors** that can **influence** this process, including the **type of information** being used and the **retrieval cues** that are present.





Theories of Memory

- Levels of Processing Theory
- Atkinson and Shiffrin Model
- Neural Network Model



Cognition

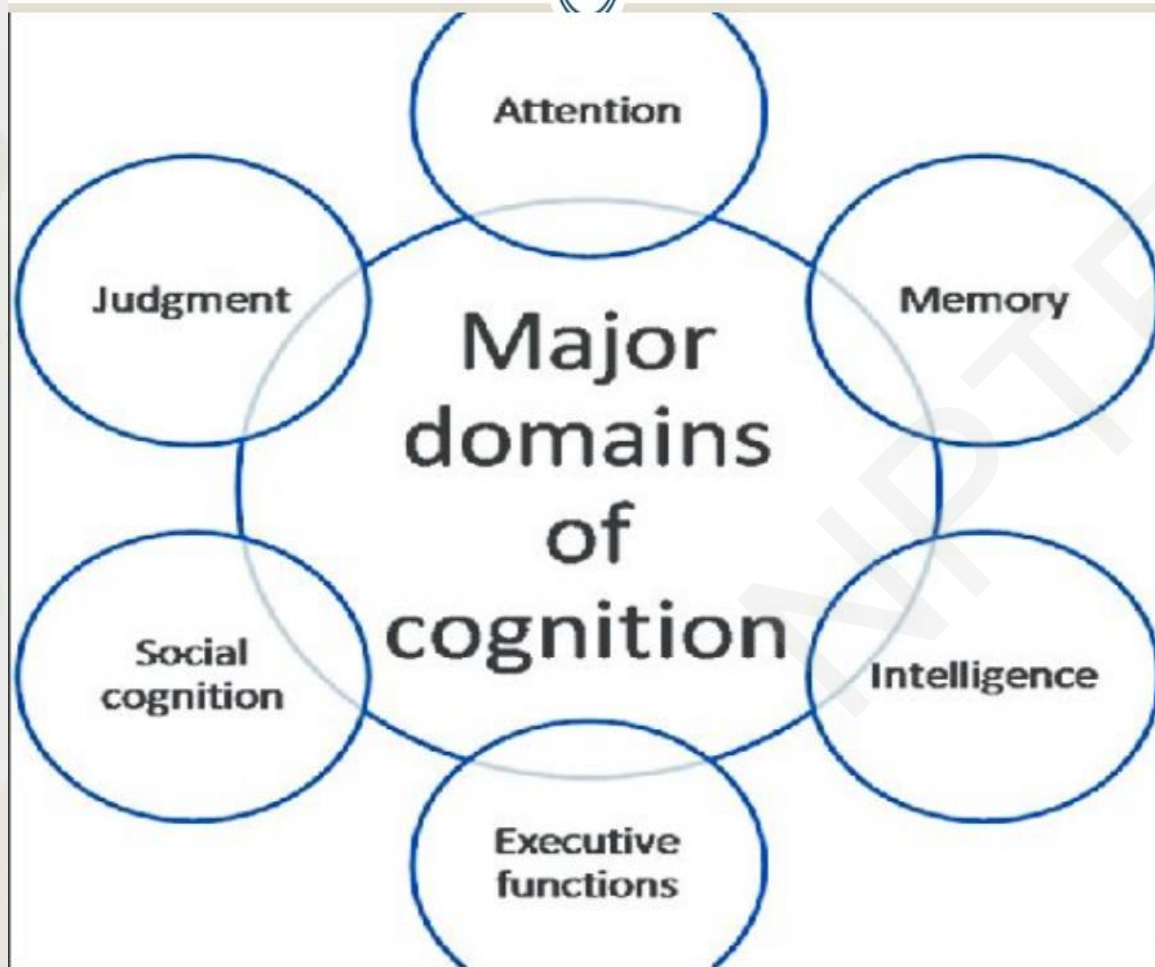


- The mental activities associated with thought, decision making, language and other higher mental processes.
- Cognition refers to a range of mental processes relating to the acquisition, storage, manipulation, and retrieval of information.
- It underpins many daily activities, in health and disease, across the age span.





Cognition

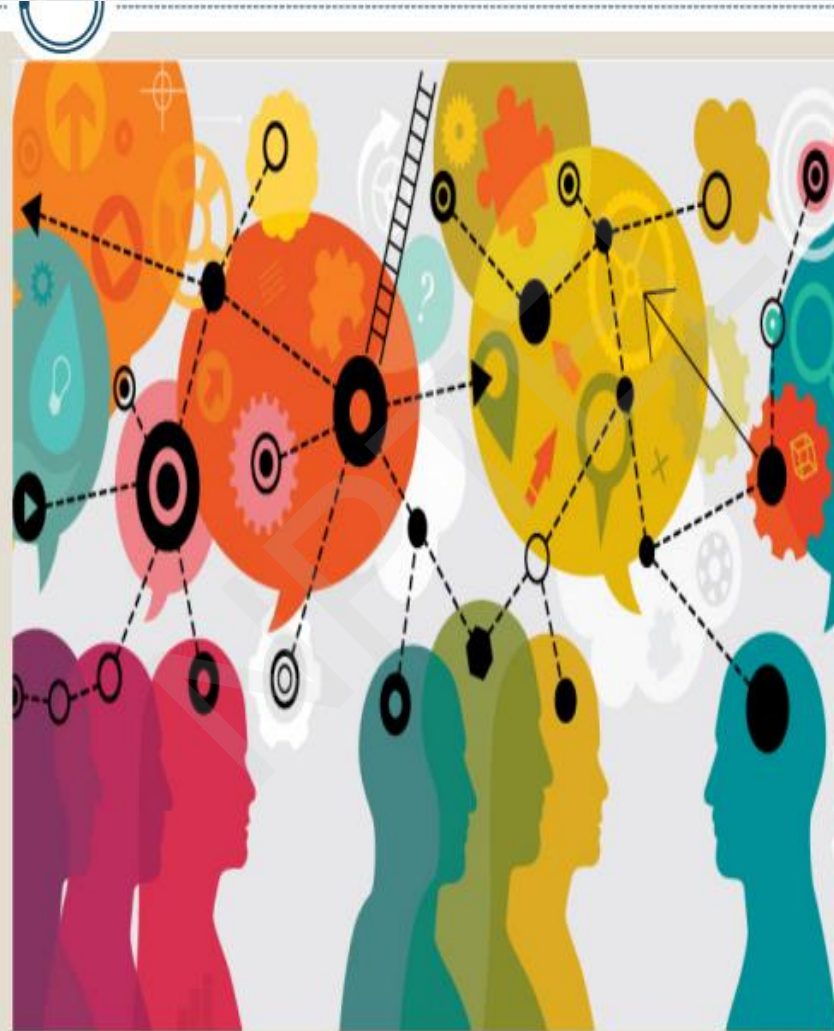




Social Cognition

The ability to develop healthy friendships and partnerships is a vital skill necessary for success in life.

For many with social cognition deficits, reading facial expressions and knowing how to respond are confusing daily challenges that pose barriers to relationships and life goals.

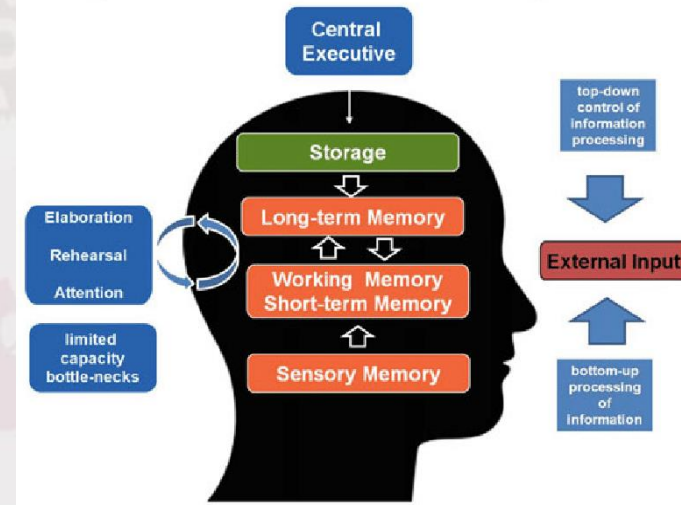


The Information Processing View of Learning

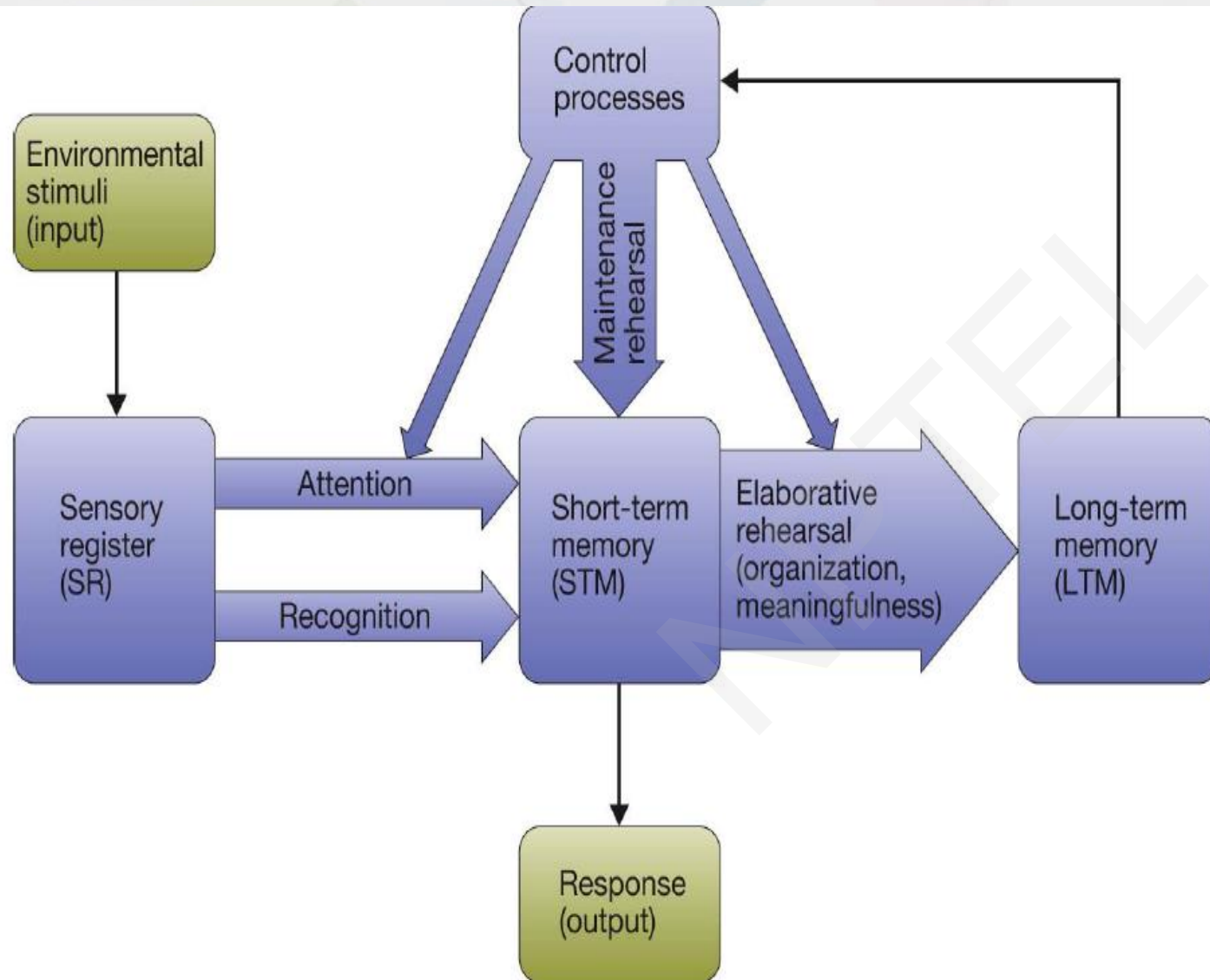
□ Assumptions

- Information is processed in steps or **stages**
- There are **limits** on how much information can be processed at each stage
- The human information processing system is **interactive**
- Information processing is **analogous to computer** processing
- Information processing is involved in all **cognitive activities** :
perceiving, rehearsing, thinking, problem solving, remembering, forgetting, and imaging

Cognitive Information Processing Model



Information Processing Model





Memory & Cognition

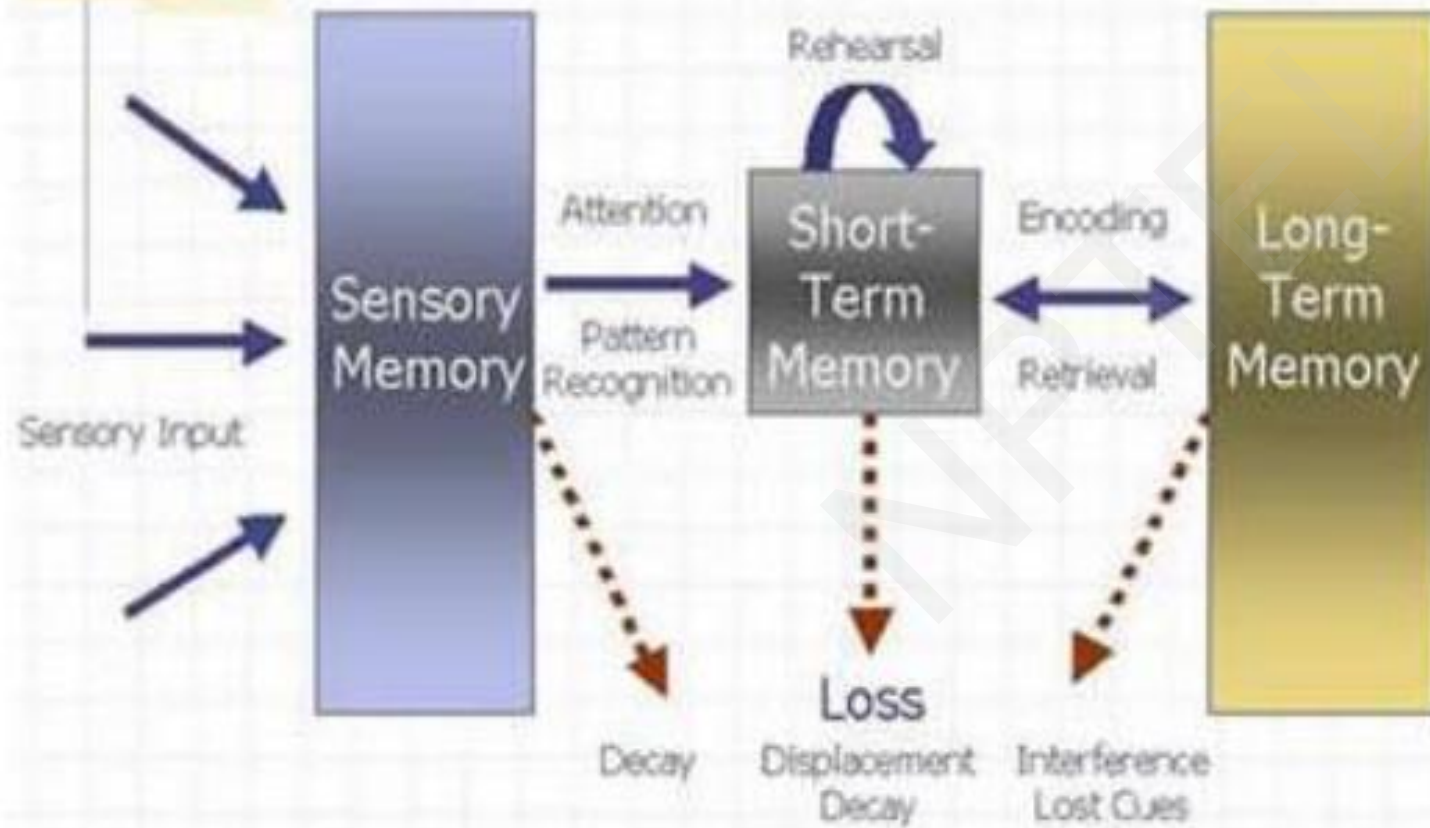
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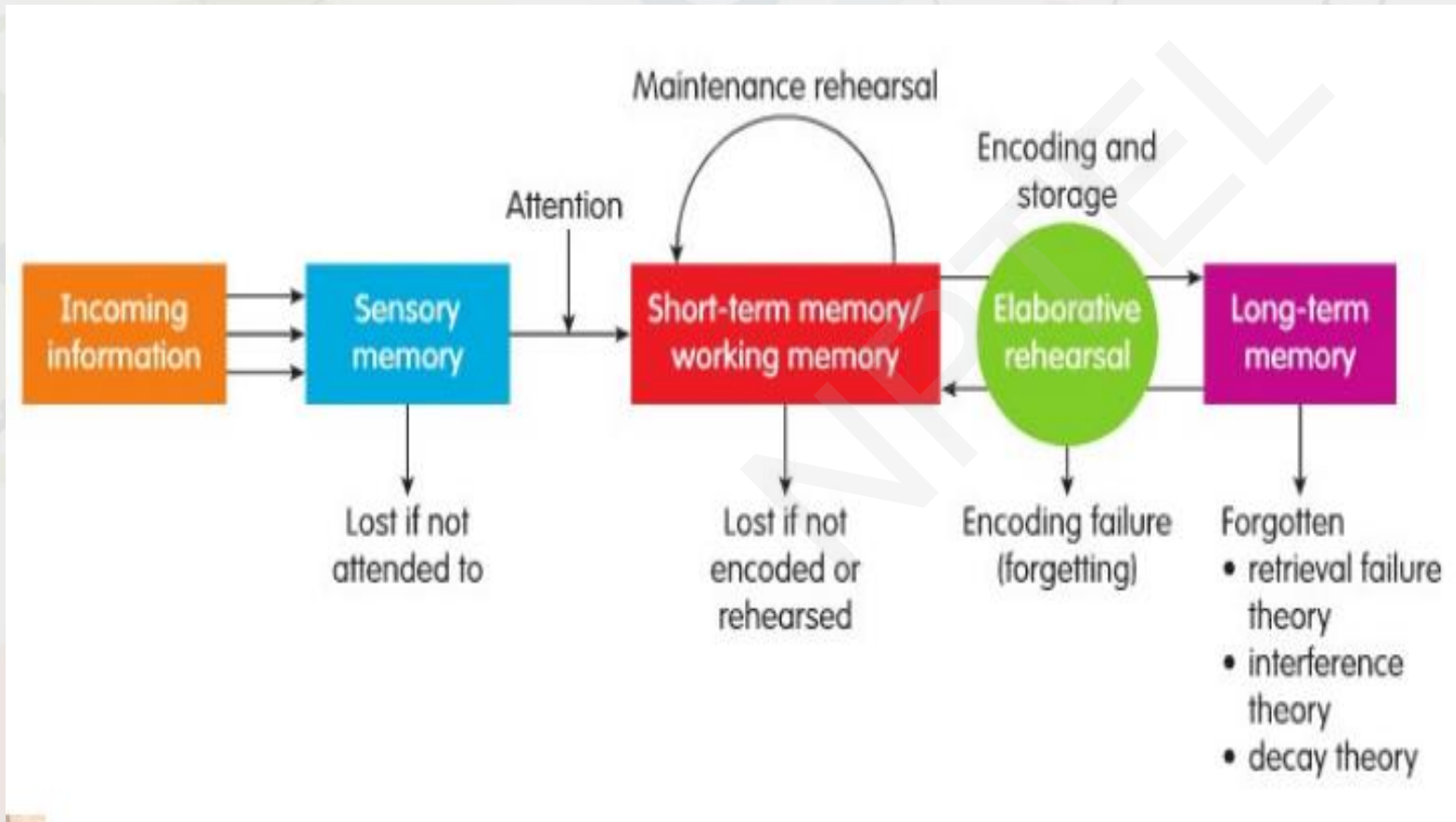


Information Processing Model





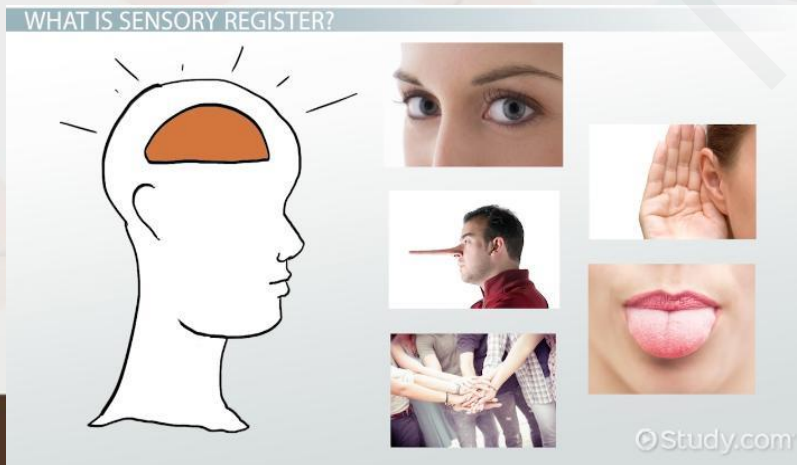
Atkinson and Shiffrin Model- Encoding- Storage-Retrieval





➤ The Sensory Register

- ✓ **Capacity** – Very large
- ✓ **Duration** – 1 to 3 seconds
- ✓ **Contents** – Raw sensory data (encoded in same form as perceived)
- ✓ The **Nature of Recognition**- Noting key features of a stimulus and relating them to already stored information
- ✓ The **Impact of Attention**- Selective focusing on a portion of the information currently stored in the sensory register
- ✓ **What we attend** to is influenced by information in long-term memory





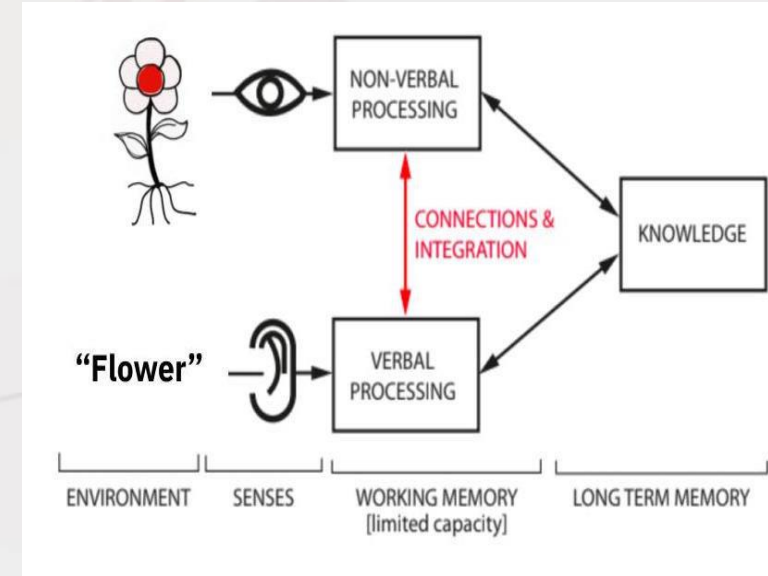
➤ Short-Term Memory

- ✓ **Capacity** – 7 +/- 2 chunks of information
- ✓ **Duration** – 20 to 30 seconds
- ✓ **Contents** – What you are currently thinking about (information from the sensory register and information from long term memory)
- ✓ **Rehearsal- Maintenance Rehearsal** (rote rehearsal, repetition): Information is repeated over and over again with no effect on long-term memory storage
- ✓ **Elaborative Rehearsal** (elaborative encoding): New information is related to knowledge already stored in long-term memory





- ✓ **Organization**- Putting interrelated pieces of information into chunks
- ✓ **Meaningfulness**- When new material can be related to information in long-term memory
- ✓ **Visual Imagery Encoding**- Generating images in your mind of objects, ideas, and actions
- ✓ **Dual coding theory**
- ✓ **Implications for Instruction**: Short-Term Memory and Its Control Processes
 - 1. **Rehearsal** prevents the quick disappearance of information from short term memory
 - 2. Most children do not begin to rehearse on their own until about age seven





- 3. **Organization** of material into chunks makes it much easier to remember
- 4. **Meaningful learning** occurs when the learner relates new information to prior ideas and experiences
- 5. **Visual imagery** is easier to recall than abstractions

➤ **Long-Term Memory**

- ✓ **Capacity** – Unlimited
- ✓ **Duration** – Permanent, long-term
- ✓ **Contents** – Schemata
- ✓ **How Information is Organized** in Long-Term Memory-
Schemata – Interrelated networks of associated ideas into which new knowledge is assimilated

Long Term Memory



- * Storage takes more time & effort
- * Unlimited capacity
- * Unlimited duration
- * Contains visual or verbal or a combination of codes
- * Retrieval may be troublesome

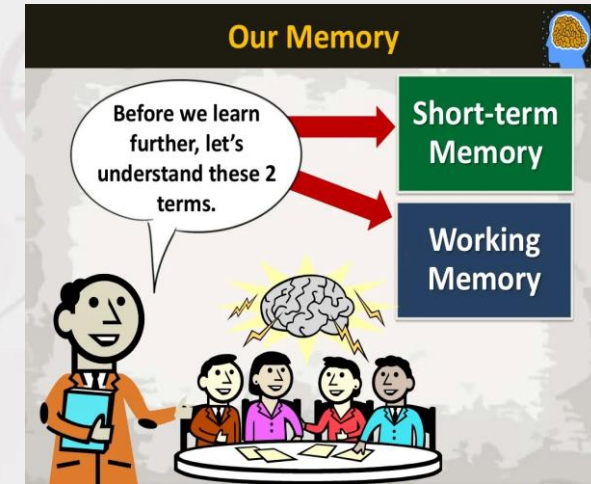
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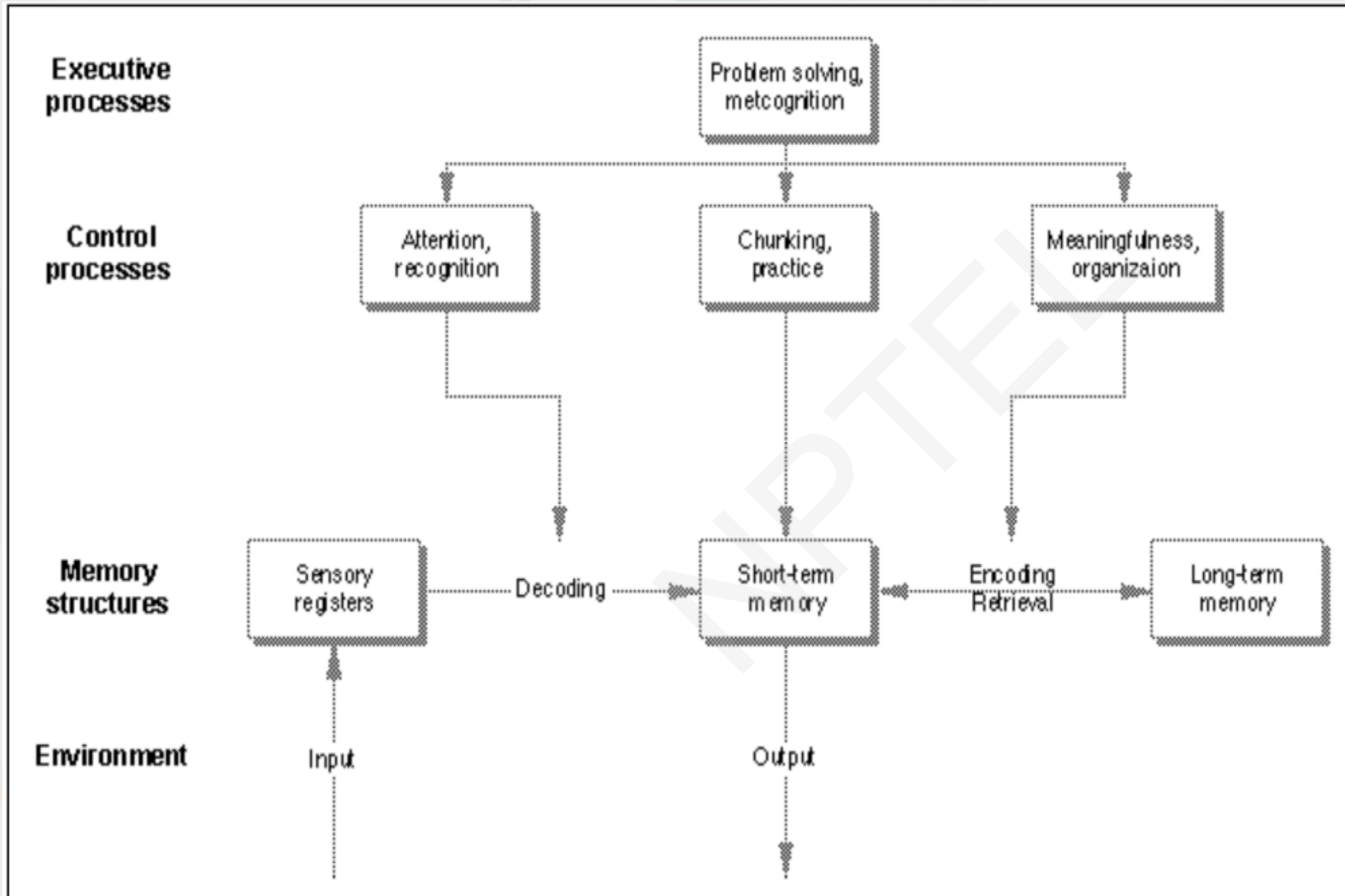


➤ How Well Do We Remember What We Learn ??

- ✓ More than 7 out of 10 studies reported less than a 20% loss of what was learned
- ✓ Subject matter that had a higher than average level of unfamiliar facts and for which students would have little relevant prior knowledge was associated with increased levels of forgetting
- ✓ Most of the forgetting of information occurred within 4 weeks after the end of a unit of instruction
- ✓ Less forgetting occurred among students who learned the material to a high level either by being required to achieve a high score, teach it to less knowledgeable students, or take advanced courses
- ✓ Less forgetting occurred in classes where students were more actively involved in learning



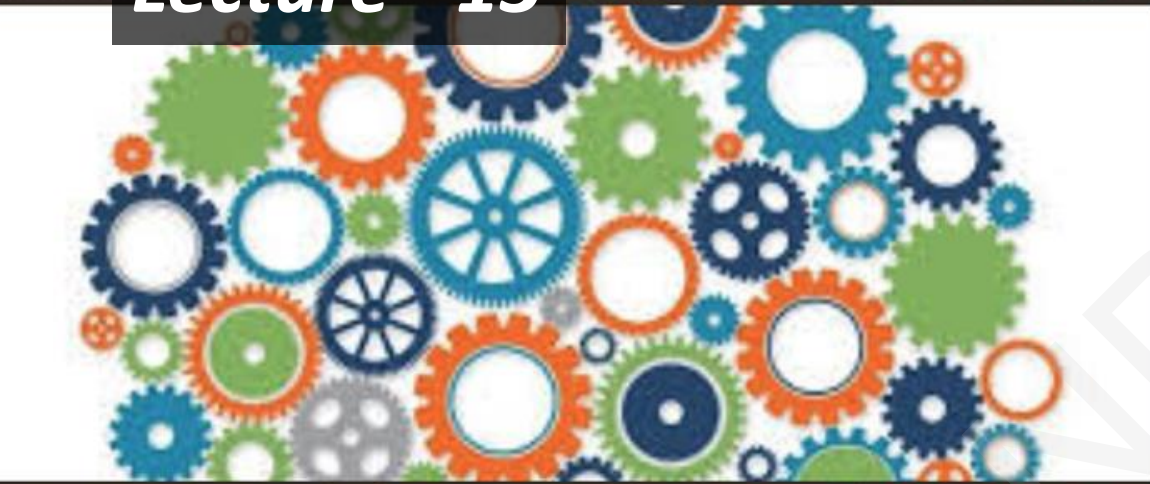
Functional Memory Processes



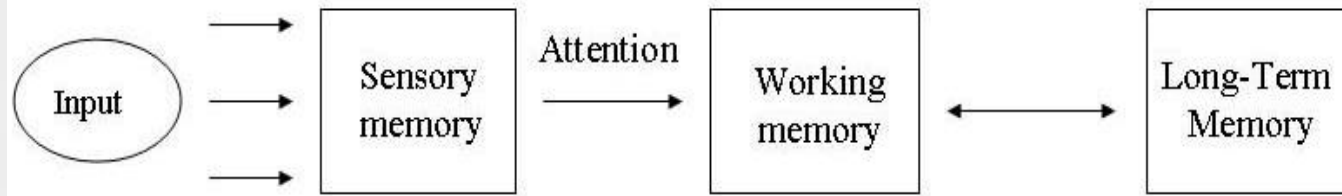


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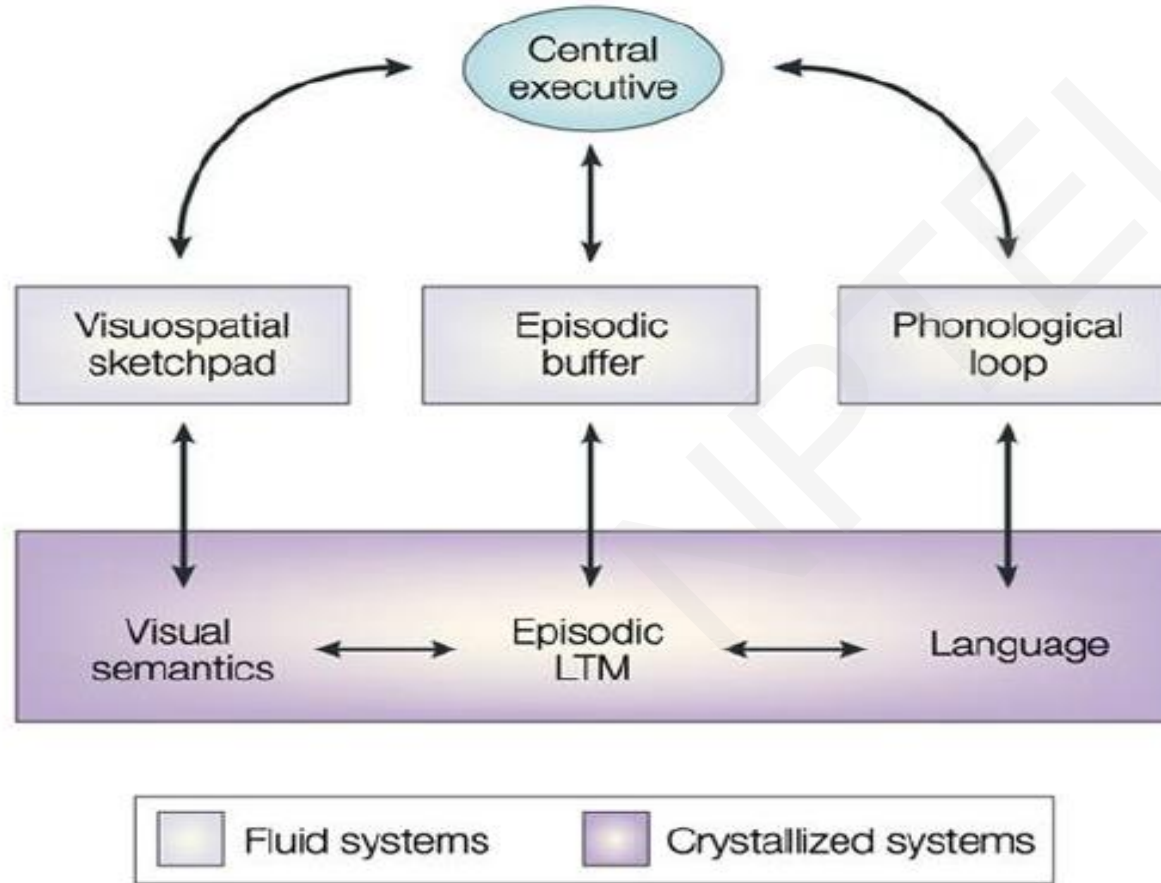
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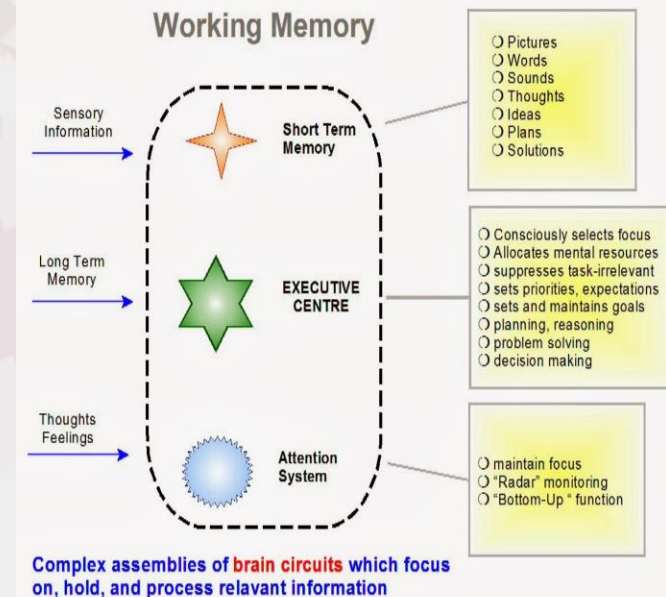
Working Memory Model Components



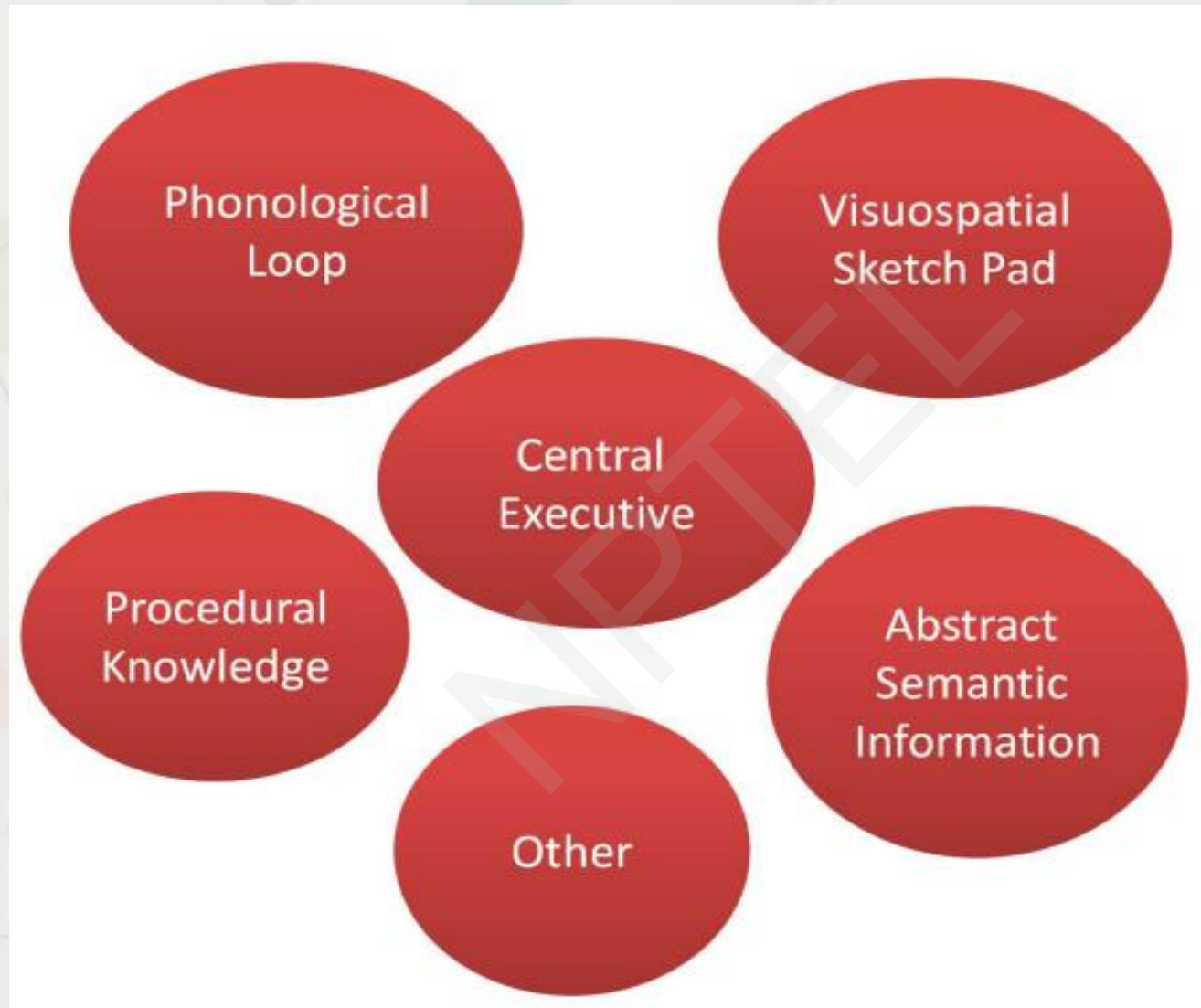
Working Memory

Assumptions:

- **1. Dual Channel**-Human information processing system consists of 02 separate channels-Acoustic/Auditory, & Visual/Pictorial/Iconic (Dual Coding Theory-Paivio & Baddley,1998)
- **2. Limited Capacity**- A limited amount of cognitive processing can take place in the verbal & visual channel at any one point of time (Cognitive Load theory-Sweller & Baddley,1998)
- **3. Active Processing**-Meaningful learning requires a substantial amount of cognitive processing to take place in verbal & visual channels(Mayer's Theory of Active L.,2002)

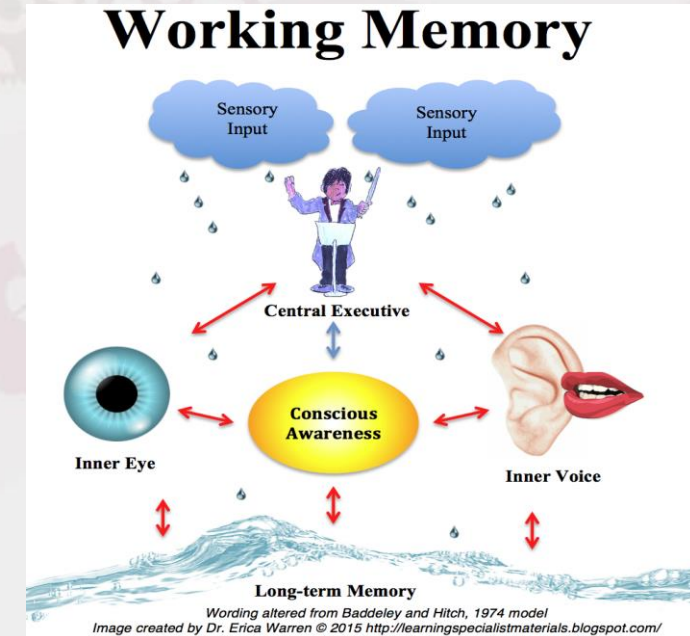


Multiple Components Model of Working Memory



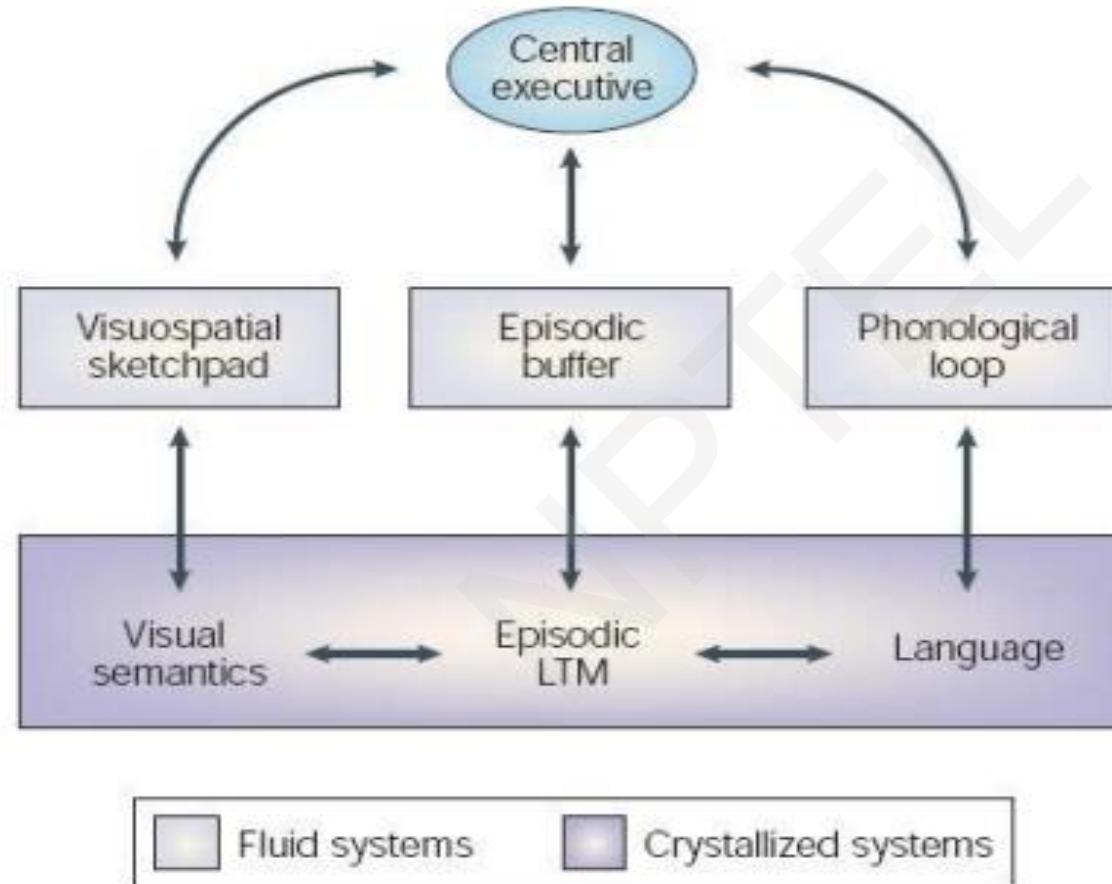
❑ Basic Principles :

- 1. Relatively independent processing system in WM for visual & auditory information
- 2. Simultaneous & mixed mode of presentations (Multimodal) can enhance WM capacity
- 3. WM is a predictor of wide range of complex cognitive tasks- planning, problem solving, decision making etc.
- 4. Central executive is responsible for controlling the overall system & performing all information processing tasks
- 5. Visuo-spatial Sketch Pad maintains & manipulates visual information
- 6. Phonological Loop stores & rehearse verbal information
- 7. Episodic Buffer-Attentional system functions as a storage structure to integrate multiple sources of information (Episodic LTM, Baddeley et al. ,2011).



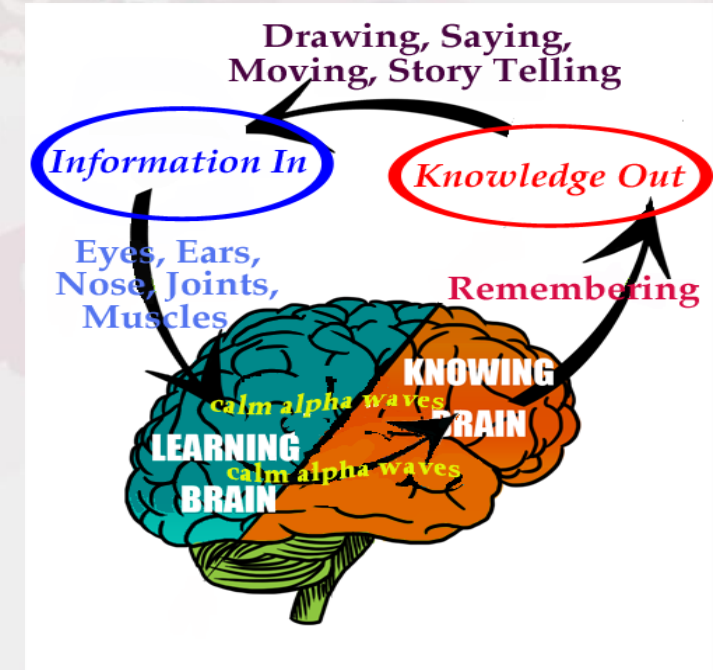
Multicomponent Model of Working Memory

(Baddeley, 2003b)



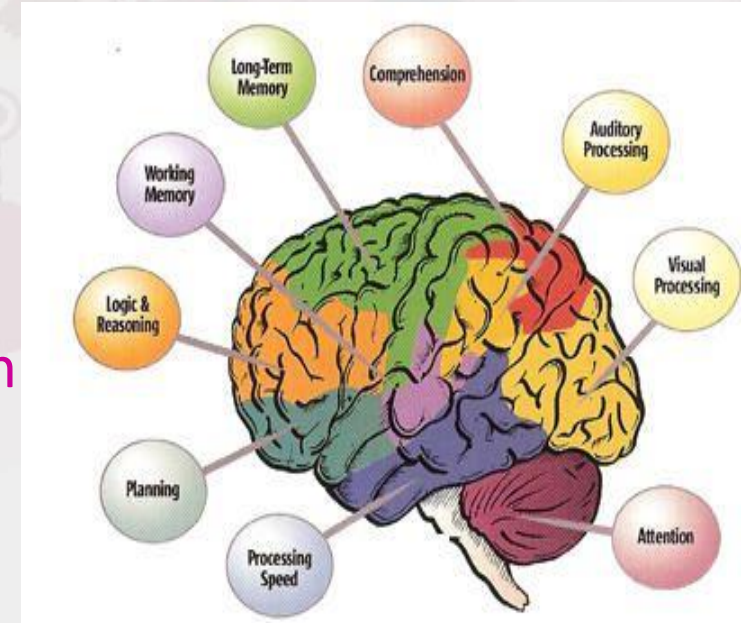


- ✓ The **Central executive** is the main component.
 - It is responsible for the **control of executive processes, including actions**;
 - the **direction of attention** to relevant information;
 - the **suppression of irrelevant information , undesired actions**;
 - the **supervision** of information **integration**;
 - the **coordination of multiple cognitive processes** to be executed in parallel;
 - and the **coordination of the sub-systems of WM**.
- ✓ The **Phonological loop** permits **to maintain**
 - **auditory information** with a **rehearsal** mechanism that prevents its rapid decay.

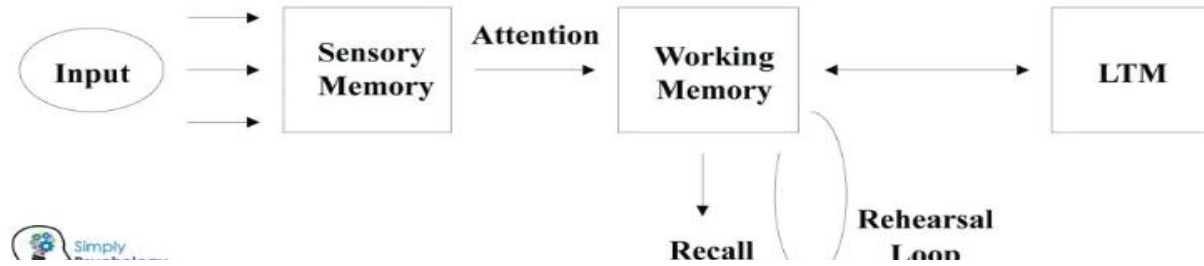




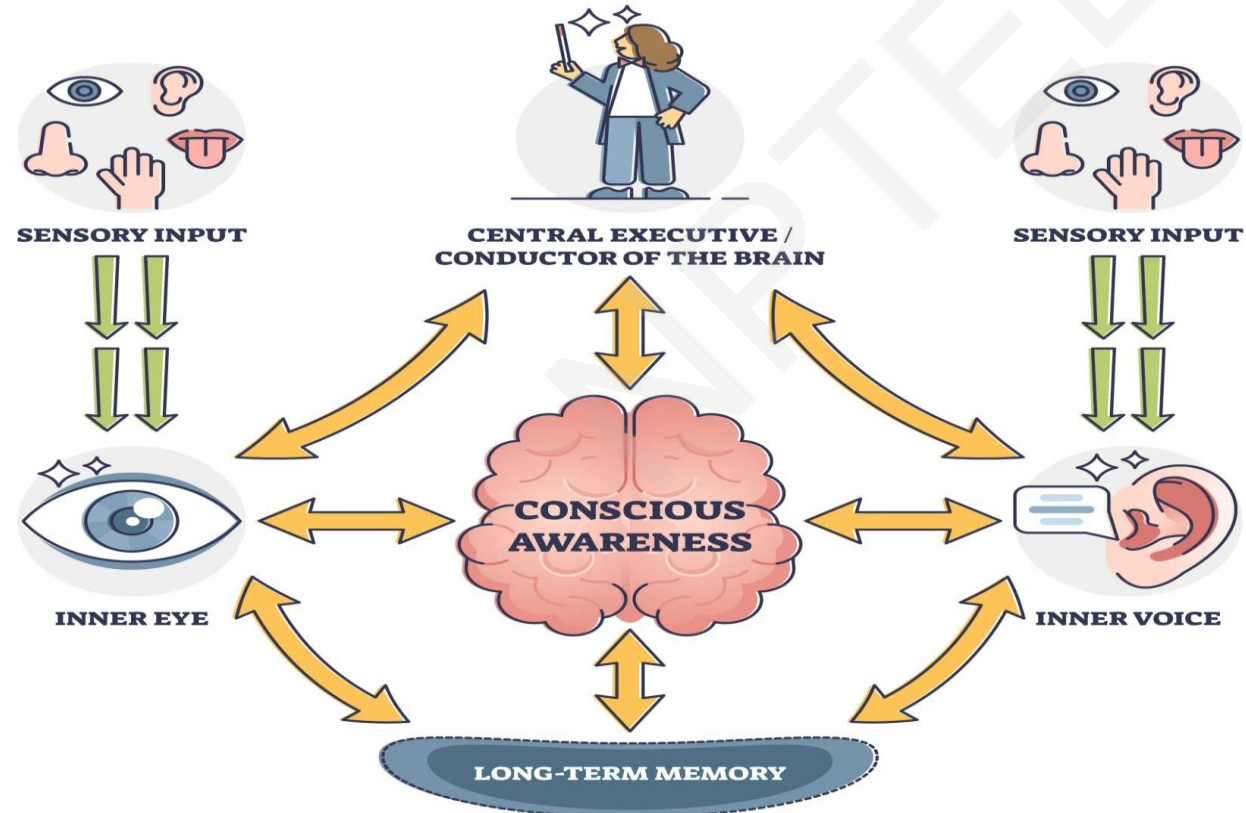
- The **Visuo-spatial sketch pad** permits to maintain and manipulate visual and spatial information. For example, we can create and navigate mental maps, form mental images and rotate them, etc.
- It is constituted of two subsystem, the former specialized for visual information, and the latter for spatial information.
- The **Episodic buffer** temporarily integrates phonological, visual, and spatial information, and possibly other forms of information (e.g., semantic information, musical information) in a unitary, episodic representation.
- Thus, it provides an interface between the sub-systems of WM and the part of LTM specialized for Episodic memory (EM) (i.e., recollection of specific events that integrate time, place, and emotions) (Tulving, 1972).



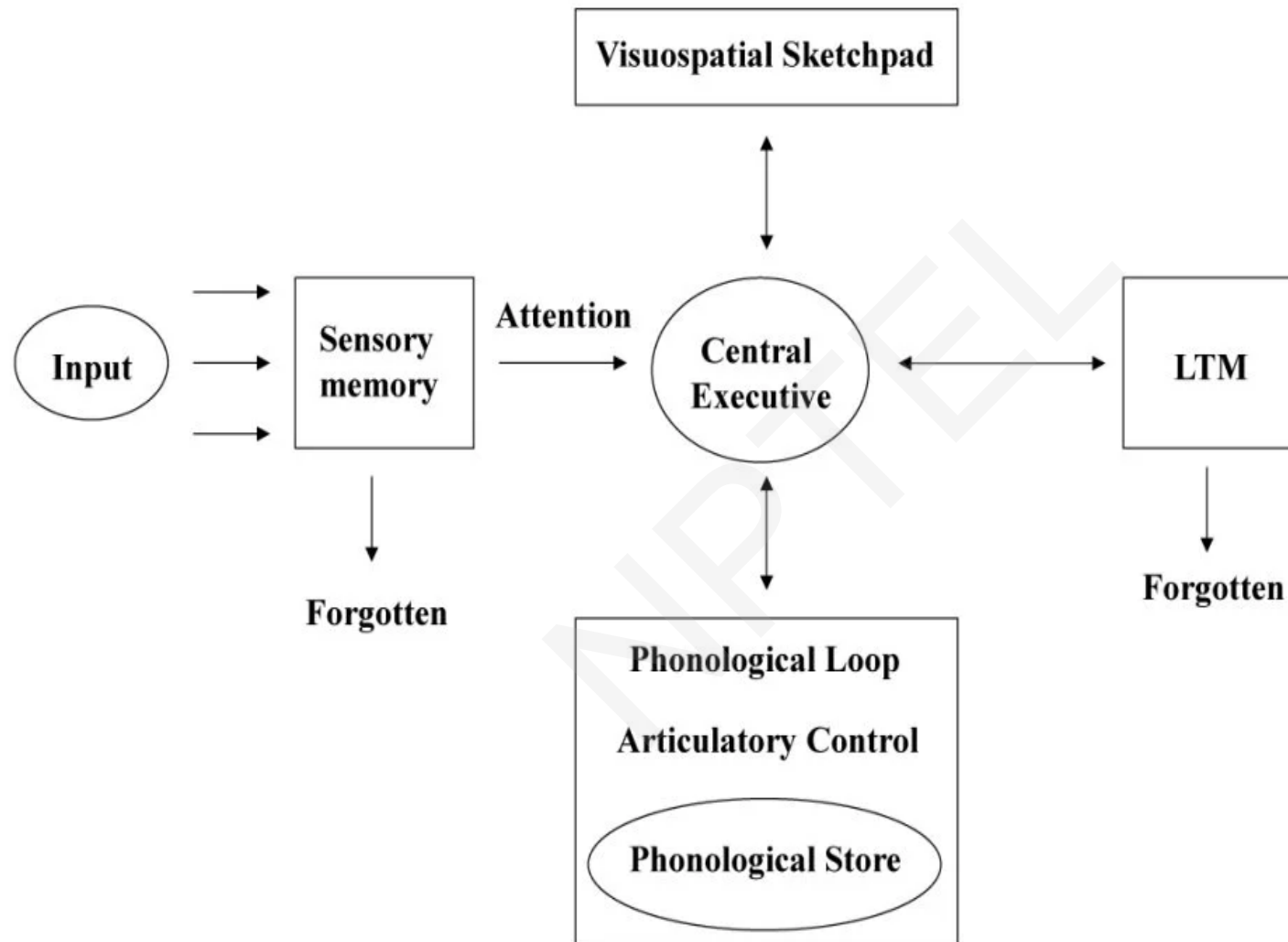
Working Memory Has Replaced STM



WORKING MEMORY



Working Memory Model (Baddeley and Hitch, 1974)

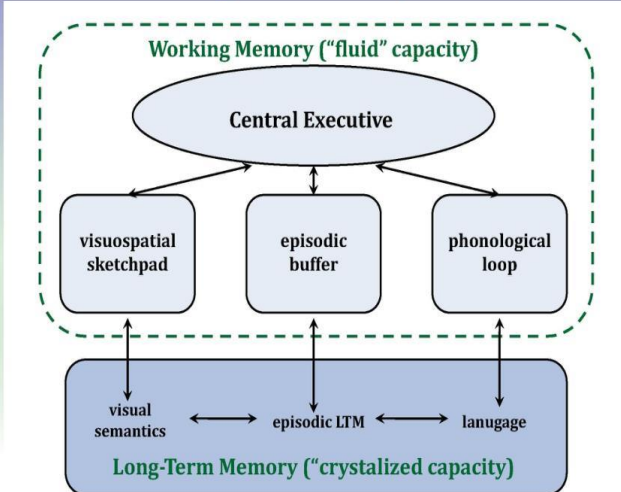




- **Summary**

- Working memory is a limited capacity store for retaining information for a brief period while performing mental operations on that information.
- Working memory is a multi-component system which includes the central executive, visuospatial sketchpad, phonological loop, and episodic buffer.
- Working memory is important for reasoning, learning and comprehension.
- Working memory theories assume that complex reasoning and learning tasks require a mental workspace to hold and manipulate information.

Working Memory Model

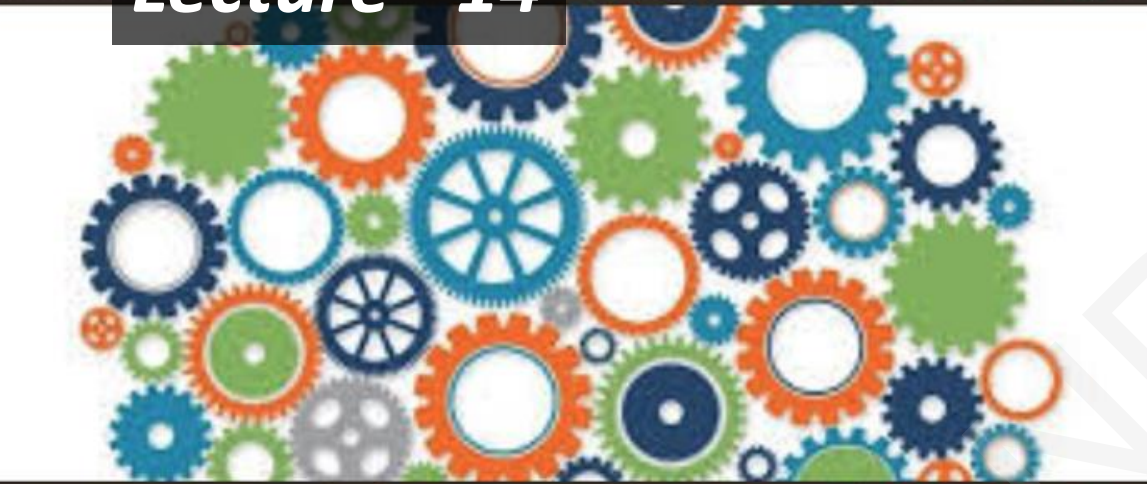


Based on Baddeley, A. D. (2000) *Trends in Cognitive Science*, 4, 417-423.
Claudia J. Stanny



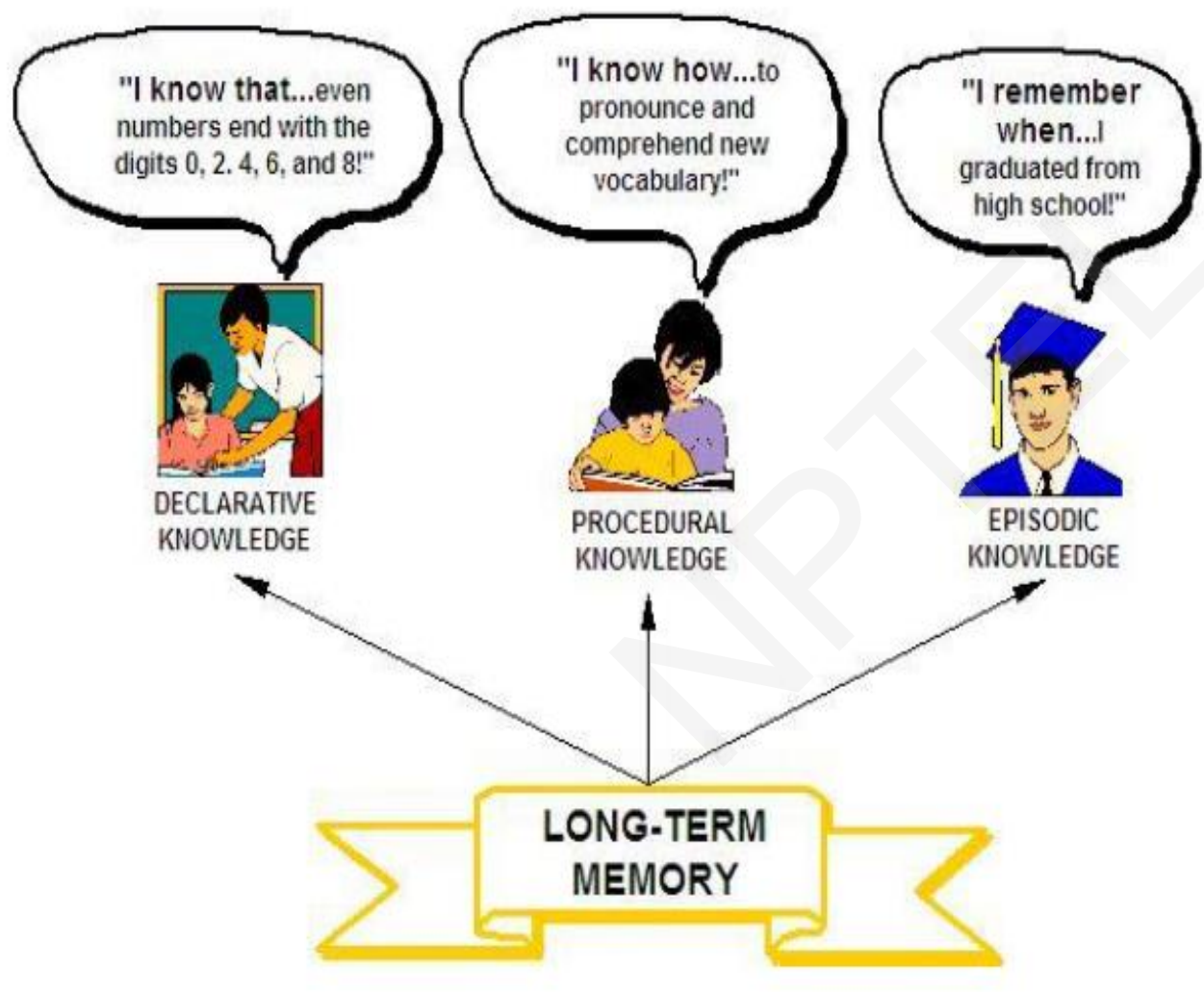
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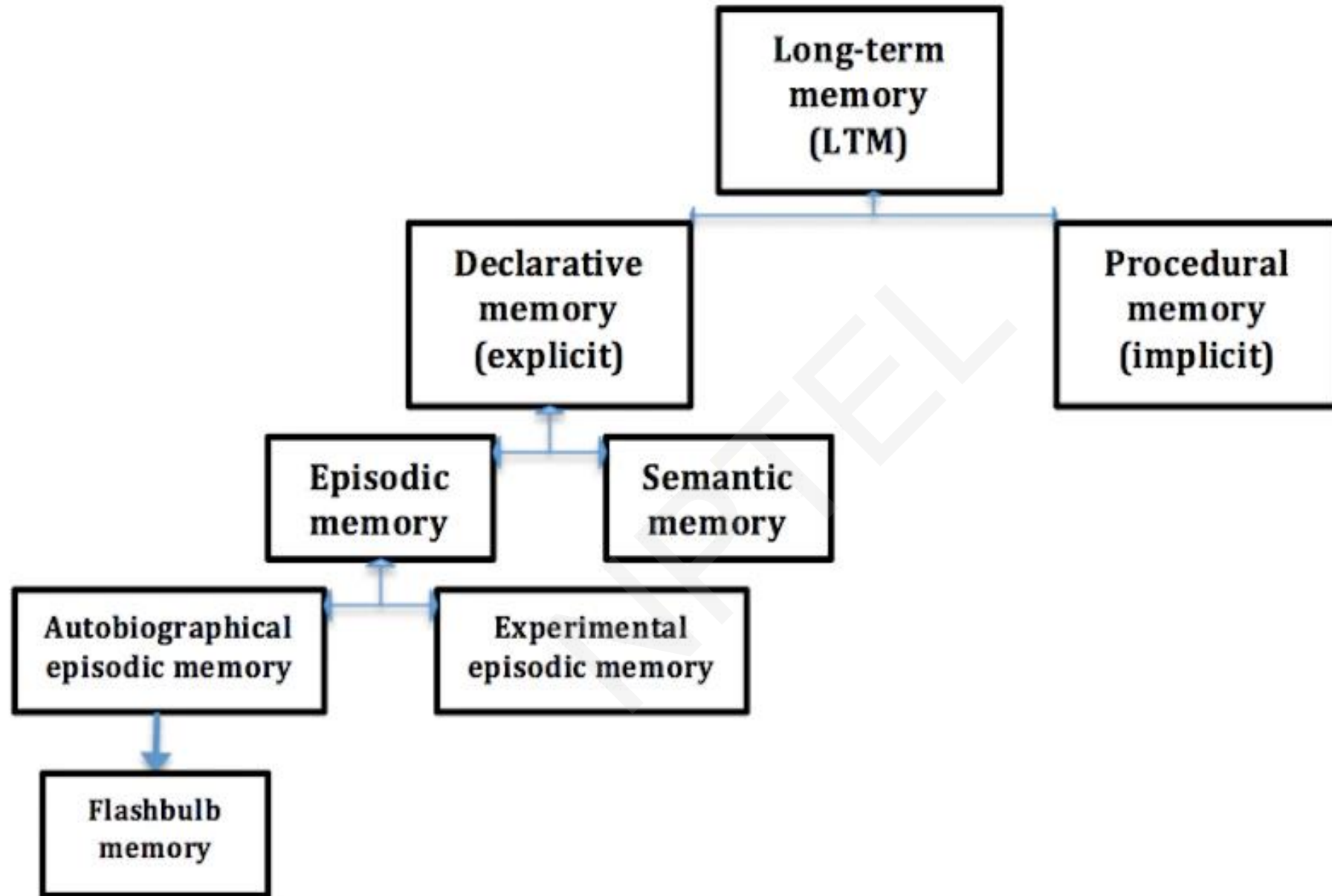
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Declarative Knowledge, Procedural Knowledge, and Episodic Knowledge in Long-Term Memory



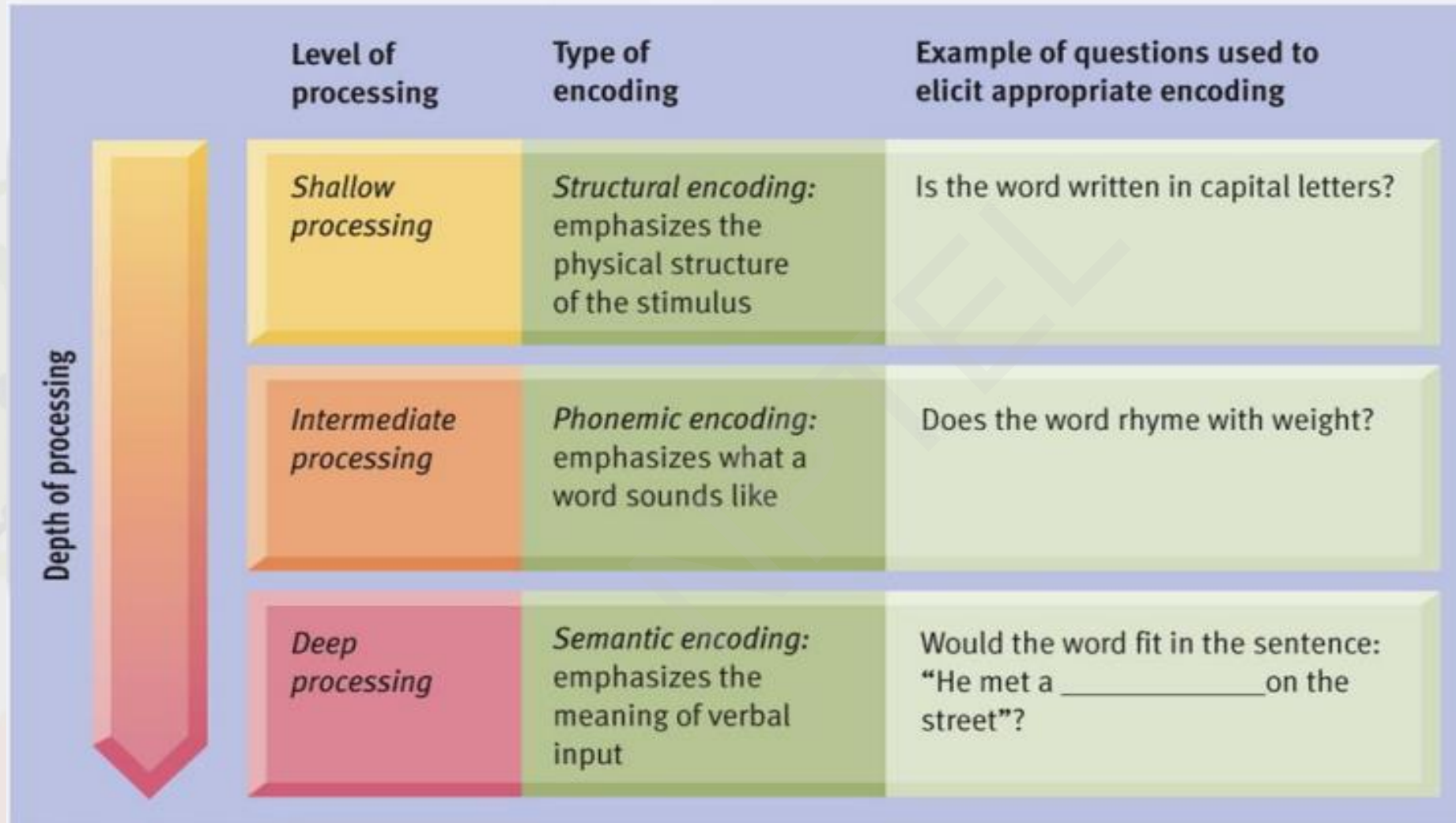




➤ Episodic, Procedural and Semantic Memory

- LTM can be split up into **declarative memories** (**explicit memories that can be inspected and recalled consciously**) and **procedural memories** (**which are implicit in that we are typically unable to consciously recall them**).
- **Declarative memory** can be further sub-categorized
 - into **episodic** and **semantic memories**,
 - **Episodic memory** refers to any events that can be reported from a person's life.
 - Episodic memory can be split further into **autobiographical episodic memory** (**memories of specific episodes of one's life**) and **experimental episodic memory** (where learning a fact [a semantic memory,
 - Like episodic memory, **semantic memory** is also **a type of 'declarative'** (**explicit, consciously recalled**) memory.
 - **Procedural memory** describes our implicit knowledge of tasks that usually do not require conscious recall to perform them.

Levels of Processing Theory- (Craik & Lockhart, 1972)

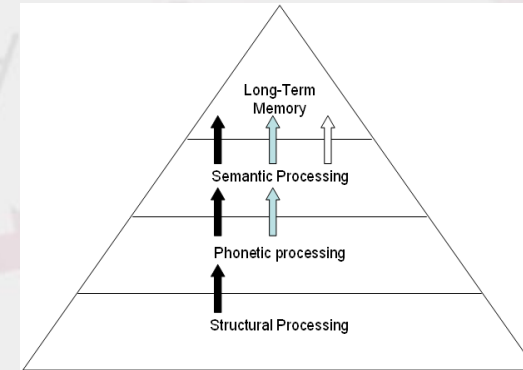
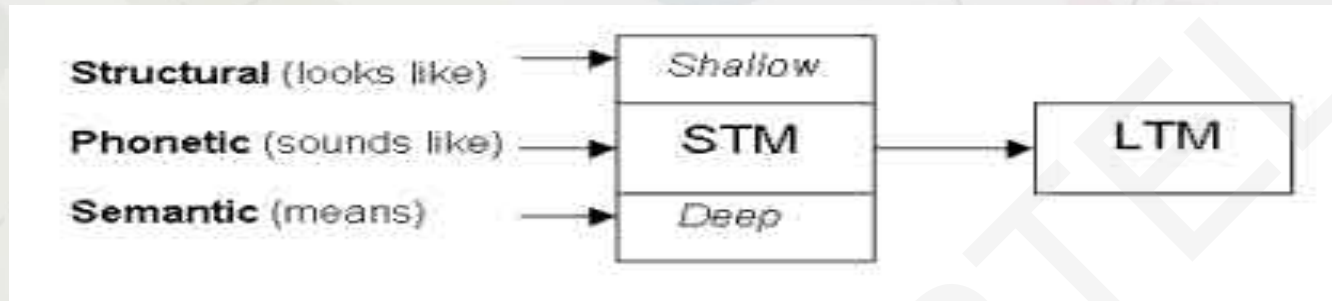


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Figure 7.3 Levels-of-processing theory



- **Levels of processing:**
- The way information is encoded affects how well it is remembered.
- The deeper the level of processing, the easier the information is to recall.



Shallow Processing only involves **maintenance rehearsal** ;

1 . **Structural processing** (appearance) which is when we encode only the physical qualities of something. **E.g. the typeface of a word or how the letters look.**

2 . **Phonemic processing** – which is when we encode its sound.

Phonemic and visually processed words involve shallow processing and less accurate recall.



- **Deep Processing**

- Deep processing involves **elaboration rehearsal** which involves a more meaningful analysis (**e.g. images, thinking, associations, etc.**) of information and leads to better recall.
- **Semantic processing**, which happens when we encode the meaning of a word and relate it to similar words with similar meaning.
- Semantically processed words involve **elaboration rehearsal** and **deep processing** which results in more accurate recall.
- Recent studies have clarified that **deeper coding produces better retention** because it is more elaborate.
- **Elaborative encoding enriches the memory representation** of an item by activating many aspects of its meaning and linking it into the pre-existing network of semantic associations.
- Latest research indicate that processing is more complex and varied than the levels of processing.



Sensory, Short-Term (Working), and Long-Term Memory

Type	Characteristics	Representation	Capacity	Duration	Cause of Forgetting
Sensory Memory	limited and passive; store information gathered from the external environment	senses (seeing, hearing, taste, feel, touch)	4 items	.5-3 seconds	decay
Short-Term Memory	active information processing: rehearsing and chunking	visual imaging and acoustic (sound) encoding	5-9 items	16-30 seconds (5-15 seconds without rehearsal)	interference
Long-Term Memory	unlimited; store information over extended periods of time (hours, days, weeks, months, years, etc.)	semantic encoding: chunking, imagery, and elaboration (knowledge: declarative, procedural, episodic, imagery, strategic, collective/ conceptual)	infinite	permanent	forgetting the retrieval pathway

Human Memory and Computer Comparison



Information Processing Perspective – Computer System

Sensory Register

Short-term Memory

Long-term Memory

INPUT DEVICES



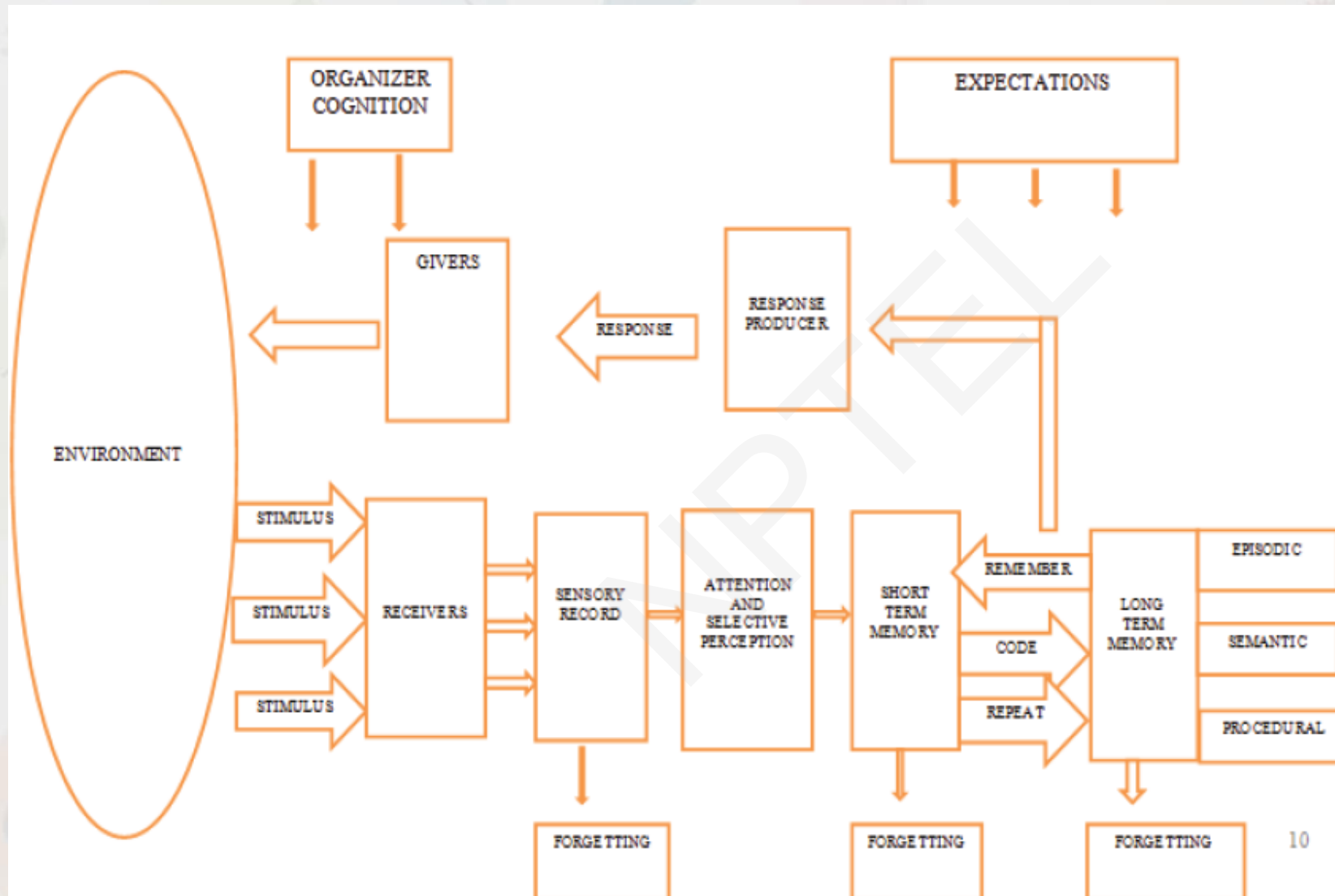
CENTRAL PROCESSING UNIT



HARD DRIVE



Human Information Processing Model





❑ Neural Network Model of Memory

- Memories are remembered as long as they influence our thoughts, feelings, and behavior at the present time.
- Memory is also one of the fundamental components of learning, our ability to acquire any type of knowledge or skills.
- The memory capacity depends on the complexity of the synapses, the sparseness of the representations, the spatial and temporal correlations between memories and the specific way memories are retrieved.
- Complexity is important when the synapses can only be modified with a limited precision, as in the case of biological synapses, and sparseness can greatly increase memory capacity and be particularly beneficial when memories are structured (correlated to each other).

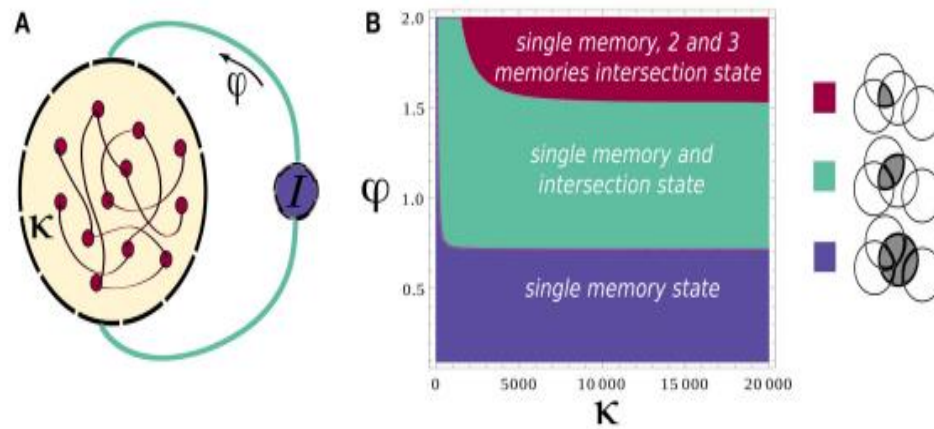


FIGURE 1 | Network architecture and Mean-field phase diagram. (A) Neurons in the network are connected through symmetric connections induced by Hebbian learning. Homeostatic control is induced by the inhibition strength determined by φ . (B) Mean-field phase diagram for the parameters K and φ . The legend illustrates different phases. Circles denote a pool of neurons encoding a particular memory. For low values of φ the single attractor solution is found, as φ is increased other solutions appear. Parameters values are according to Table 1.

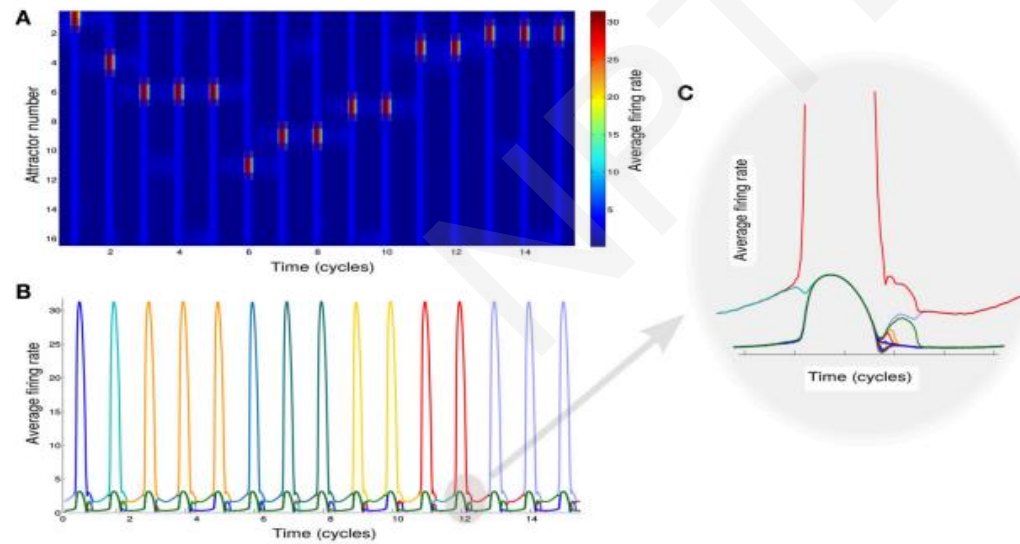


FIGURE 2 | Neural network activity. (A) Activity of the attractors in the network. Different rows correspond to the average firing rate of different memories for 15 cycles of oscillation of φ . (B) Activity of the attractors in the network. Each colored line corresponds to the average firing rate of a different memory. (C) Details of the neuronal dynamics.



- **Research finding- Example**

- **Model for memory retrieval based on a Hopfield neural network:**

- Here the transition between items are determined by similarities in their long-term memory representations.
- Meanfield analysis of the model reveals stable states of the network corresponding ,
- (1) to single memory representations and
- (2) intersection between memory representations.
- It shows that oscillating feedback inhibition in the presence of noise induces transitions between these states triggering the retrieval of different memories.
- The network dynamics qualitatively predicts the distribution of time intervals required to recall new memory items observed in experiments.
- It shows that items having larger number of neurons in their representation are statistically easier to recall and reveals possible bottlenecks in our ability of retrieving memories.
- This neural network model of information retrieval broadly compatible with experimental observations and is consistent with the graphical model ([Romani et al., 2013](#)).

❑ Cognitive Load

- Cognitive architecture: memory and schemas
- Schema theory-, knowledge is stored in LTM in schemata
- Schemata categorize information elements according to how they will be used
- A schema can hold a huge amount of information, yet is processed as a single unit in working memory
- Schemata can integrate information elements and production rules and become automated, thus requiring less storage and controlled processing
- Skilled performance consists of building increasing numbers of increasingly complex schemas by combining elements in an inductive/sequential manner





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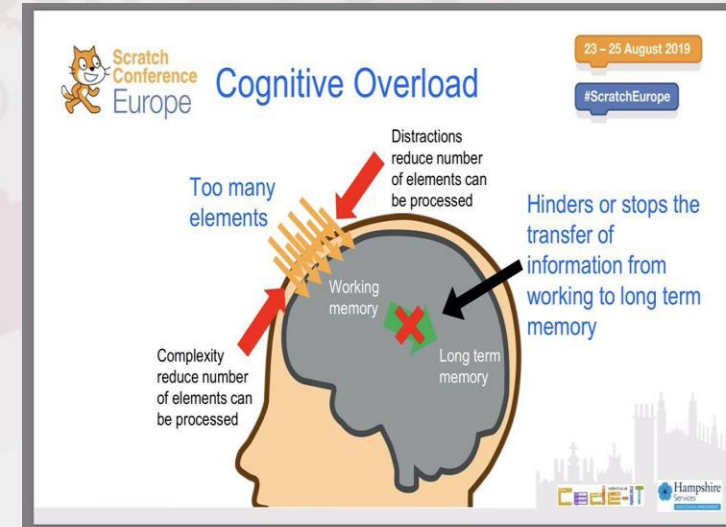
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- Schemas can also reduce working-memory load
- Schema construction aids the storage and organization of information in long-term memory and reduces working memory load
- Cognitive Load Theory (CLT) assumes a limited working memory connected to an unlimited long-term memory (Baddeley, 1986)
- Both causal and assessment factors affect CL
- Causal factors can be characteristics of the subject (e.g. cognitive abilities), the task (e.g. task complexity), the environment (e.g. noise), and their mutual relations



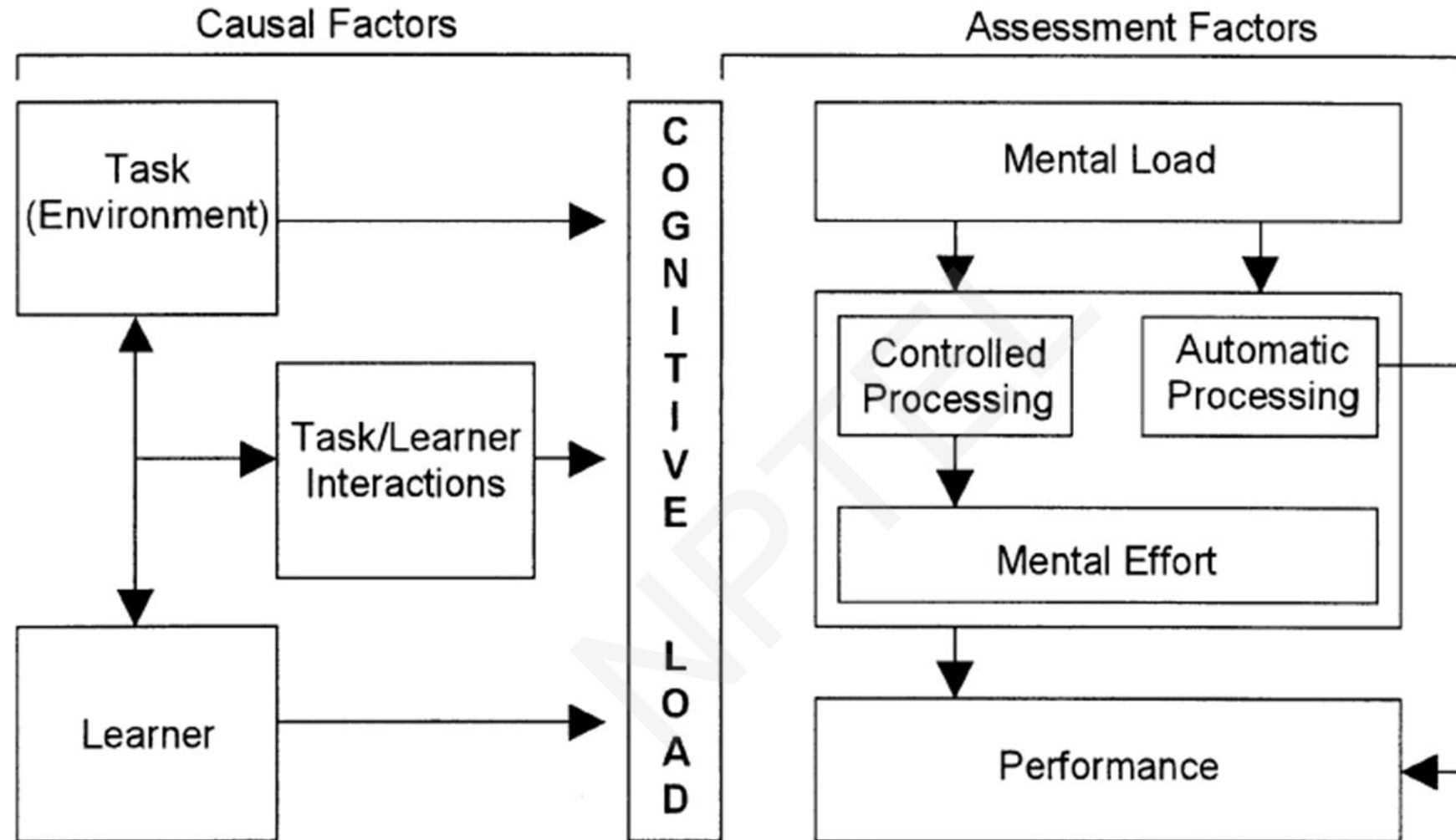


Fig. 1. Factors determining the level of cognitive load.



- Assessment factors include mental load, mental effort, and performance as the three measurable dimensions of CL
- Mental load is the portion of CL that is imposed exclusively by the task and environmental demands
- Mental effort refers to the cognitive capacity actually allocated to the task
- The subject's performance, finally, is a reflection of mental load, mental effort, and the aforementioned causal factors
- Working memory load is affected by the inherent nature of the material (intrinsic CL) and by the manner in which the material is presented (extraneous and germane CL)

3 Types of Cognitive Load

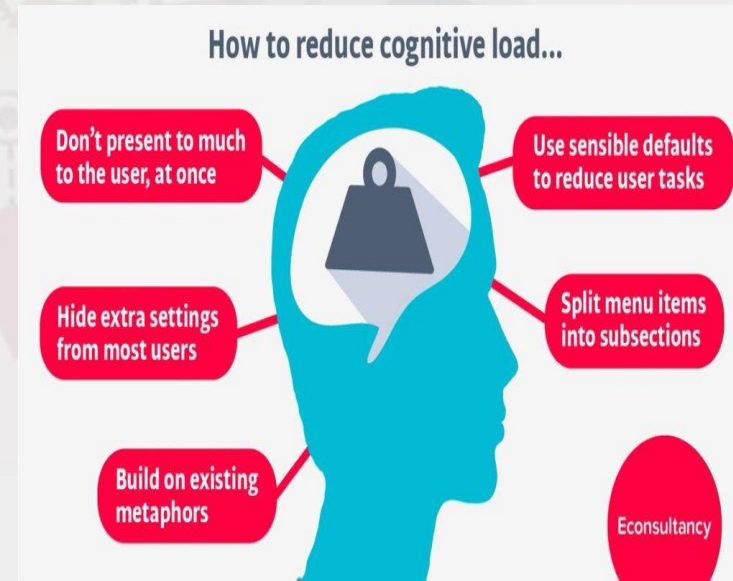
- Intrinsic Load
 - It is the memory required by the thinking task at a given time.
- Extraneous Load
 - Results from the techniques in which the to-be-learned information is presented.
- Germane Load
 - Is the load that helps building new complex schema in a successive manner helping the learner to move from novice to expert.



- **Germane CL** is required for the construction and storage of schemata into long-term memory
- **Intrinsic CL**, which is the portion of load that is imposed by the intrinsic characteristics of the task or subject matter
- Conventional instructions tend to impose an **Extraneous CL** on working memory, whereas learning something requires shifting from extraneous to germane CL
- **Instructional design** -unused working memory capacity because of a **low intrinsic CL** by the instructional materials or **low extraneous CL** due to appropriate instructional procedures ,may further improve L. by **encouraging learners to engage in conscious cognitive processing relevant for schema construction**
- **Total CL of the instructional design-** (intrinsic CL +extraneous CL+ germane CL) is within working memory limits.



- Effective ways of reducing Cognitive Load:
- 1. **One concept per page**-Presenting one key learning concept per page will reduce cognitive load
- 2. **Remove non-relevant items**-removing media elements not relevant to learning
- 3. **Clear, concise writing style**-Remove excess words and statements that don't relate directly to the topic
- 4. **Information access- Clever chunking** of content with smaller, bite-sized pieces of information, pop-ups or click-to-reveal mechanisms to allow learners to drill down and discover content one step at a time

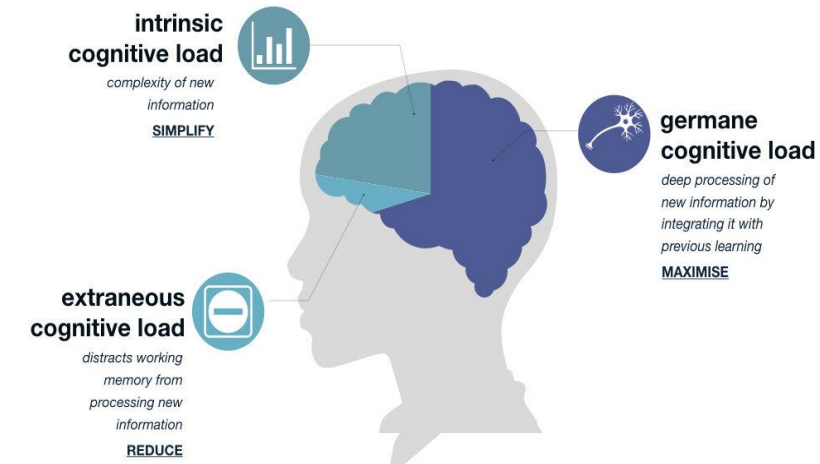




- 5. **Consolidate key learning**-Consolidating key learning points at the end of each topic or module will increase knowledge retention
- 6. **Use consistent structure**-In eLearning, learners will not need to continuously readjust to the experience, leaving more cognitive load available for information assimilation
- 7. **Link back to learning objectives**-create strong links back to learning objectives to reinforce key learning, will make it easier for learners to grasp key learning points, thus reducing brain power required to create these links

cognitive load

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- **How to Protect Our Memory-**

- **Avoid stress:** Research has found that stress can have detrimental effects on areas of the brain associated with memory, including the hippocampus.
- **Avoid drugs, alcohol, and other neurotoxins:** Drug use and excessive alcohol consumption have been linked to the deterioration of synapses (the connections between neurons). Exposure to dangerous chemicals such as heavy metals and pesticides can also have detrimental effects on the brain.
- **Get enough exercise:** Regular physical activity helps improve oxygenation of the brain, which is vital for synaptic formation and growth.
- **Stimulate your brain:** When it comes to memory, there is a lot of truth to the old adage of "use it or lose it." Researchers have found that people who have more mentally stimulating jobs are less likely to develop dementia.



- **Maintain a sense of self-efficacy:** Having a strong sense of self-efficacy has been associated with maintaining good memory abilities during old age.
- **Self-efficacy-** refers to the sense of control that people have over their own lives and destiny. A strong sense of self-efficacy has also been linked to lowered stress levels.
- **How to Improve Memory-**
- **Write it down:** The act of writing with a pen and paper helps implant the memory into your brain—and can also serve as a reminder or reference later on.¹⁰
- **Attach meaning to it:** You can remember something more easily if you attach meaning to it. For instance, if you associate a person you just meet with someone you already know, you may be able to remember their name better.
- **Repeat it:** Repetition helps the memory become encoded beyond your short-term memory.



- **Group it:** Information that is categorized becomes easier to remember and recall.
- **Test yourself:** While it may seem like studying and rehearsing information is the best way to ensure that you will remember it, researchers have found that being tested on information is actually one of the best ways to improve recall.
- **Take a mental picture:** Systematically trying to make a mental note of things you often forget (such as where you left your car keys) can help you remember things better.
- **Get enough rest:** Research has also found that sleep plays a critical role in learning and the formation of new memories.
- **Use memorization techniques:** Rehearsing information, employing mnemonics, and other memorization strategies can help combat minor memory problems.





THANK YOU!!